

BBO capstone project presentation template

Develop a concise presentation that captures:

1. An overview of your BBO approach

- Describe what you are trying to achieve or discover through the BBO capstone project. Focus on the core objective of your optimisation process, using clear and accessible language.
- Briefly describe your overall strategy and how your process fits together.

My objective

I am trying to find high-value inputs for eight unknown functions without knowing their formulas. Each function has between two and eight input variables, and I can submit only one point per function in each round.

My approach

I start with the supplied observations, add the results from each round and use a Gaussian process model to estimate which new points might be useful. I balance promising areas with areas where the model is less certain.

The process

Review the data, compare possible points, submit one query per function, record the returned values and update the model. From round 11, I also group similar inputs and give a small preference to the group with the best average result.

How I judge progress

I compare the best output found for each function with its initial best, while also checking the latest result. A good prediction or one successful query is not proof that I have found the best possible input.

Current evidence: 175 initial observations plus 104 query records across thirteen completed rounds, for 279 input-output records.

2. How your strategy has evolved

- Describe the key changes in your approach since your early rounds.
- What influenced these changes – data trends, feedback, model performance or intuition?
- Identify which principles or heuristics now guide your query decisions.

Starting point

The initial Function 1 script compared several model settings using expected improvement: choosing a point for its potential to beat the best known result. By Week 2, I used GP-UCB, which also gives weight to uncertainty.

More systematic searching

From round 3, I made small exploration adjustments when recent results improved or worsened. Candidate budgets were larger for functions with more variables. Fixed seeds and saved query files made the process easier to repeat.

Comparing alternatives

I tested small neural networks on Functions 5, 7 and 8 as a separate exercise. Their performance varied with the settings and limited data. The submitted-query scripts continued using Gaussian processes; the network experiment was not a replacement query policy.

Rounds 11 and 12

I added input-space clustering to favour promising local groups while retaining uncertainty-driven search. The current cluster-aware code uses dimension-based exploration rather than the earlier recent-result adjustment.

What guides me now

Build on all available evidence, compare results within each function, refine promising regions and avoid treating one high value as a reliable pattern. Model warnings and uneven outcomes remind me that more computation does not replace better coverage.

3. Patterns, data and insights

- Highlight the most meaningful trends you've observed across your data points or model evaluations.
- Identify which variables or behaviours seem to influence results the most.
- Explain how these observations have shaped your understanding of the search process.

Results through round 13. Higher is better within each function; output scales differ, so values should not be compared across functions.

Function	Initial best	Best observed	Final round 13
F1	~0	0.00000	0.00000
F2	0.61121	0.64245	0.61859
F3	-0.03484	-0.03484	-0.08548
F4	-4.02554	0.66151	0.57686
F5	1,088.85962	8,662.40500	8,662.40500
F6	-0.71426	-0.32829	-0.32829
F7	1.36497	2.12521	1.88081
F8	9.59848	9.96927	9.96927

Meaningful patterns

The best observed value improved over the initial best for seven functions; F3 did not improve. F5 rose from about 1089 initially to 8662.41. F2 reached 0.64245 in round 12, while the final F2 result was lower at 0.61859. F4 recovered from poor boundary results and finished at 0.57686.

What seems to matter

Location and combinations of inputs appear important. Because I usually change several variables together, I cannot isolate individual causes. F1's tiny outputs give little useful signal, and the larger input spaces remain sparsely covered.

What I have learned

I need to look at both improvement and setbacks, not just the largest score. Grouping and PCA-style thinking can help summarise patterns, but PCA has not been applied here and input variation alone does not establish which variables drive high outputs.

4. Decision-making and iteration

- Explain how you balance exploration and exploitation.
- Describe one or two examples of strategic decisions. What worked, what didn't, and why?
- Reflect briefly on how you handle uncertainty or adapt when results don't match expectations.

Balancing the search

Exploration tests uncertain areas; exploitation tests near promising results. My model uses both, with a small bonus for a promising input group. The same balance is not equally useful for every function.

What worked: Function 5

The best observed value rose from about 1089 initially to 6814 in round 12, then to 8662.41 in the final round. These results came from points near the upper input bounds. This is a strong observed pattern, but it does not prove the global optimum or isolate which input caused the gain.

What worked: Function 4

Two earlier boundary queries returned below -34. Moving back near the central observations produced positive results, including 0.57686 in the final round. This was a useful correction after the model's boundary predictions performed poorly.

Mixed evidence and uncertainty

The final result was below the previous best for Functions 2, 3, 4 and 7, although F4 remained positive. A promising query can still fall short of the best result. Identical repeated inputs also differed for Functions 2, 3 and 6, so observed variability and record uncertainty deserve attention.

5. Next steps and reflection

- Outline your planned actions or experiments to improve the model's performance.
- Include two or three sentences connecting your work to the broader ML landscape.
- Add a short statement on how you'd communicate your results to a non-technical audience or stakeholder.

Project close-out

The final query round is complete. The next steps are to preserve the raw records, verify repeated-query provenance and report the best observed result separately for each function. The search did not establish global optima.

What I would improve

Record prediction and uncertainty diagnostics consistently, compare clustered and non-clustered policies under the same query budget, and reserve some queries for deliberate space coverage. Repeated inputs should be interpreted only after their source is confirmed.

Connection to wider ML

This project shares the challenge of active learning: deciding which new observation is worth its cost. Like other ML work, its results depend on data coverage and model assumptions, not just model complexity. PCA highlights the value of finding patterns and reducing redundancy, but a low-variation variable can still be important to the objective.

Explaining the results

For a non-technical stakeholder, I would say: 'I used each result to make the next experiment more informed. Some functions improved, others remained uncertain, and the best values found are not guaranteed to be the best possible.'

Evidence: repository scripts, initial arrays, weekly logs, cluster diagnostics and reflections through the final round. The datasheet and model card document the data and method limits.